

Nusselt Number Correlations for a Plate Heat Exchanger

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Abstract

Plate heat exchangers offer higher heat transfer area to volume ratios and thermal effectiveness compared to other heat exchangers. This requires more precision in understanding overall heat transfer coefficients and thermal energy. This experiment verifies Nusselt's number correlation by comparing the derived overall heat transfer coefficient to the measured value. The coefficients of the model were determined to be $C_1 = 1.7868$ and $C_2 = 0.6853$. All values were under a 95% confidence interval, verifying the theoretical model was significantly precise. The energy balance of heat transfer only contained 88.9% of values within $\pm 25\%$ of the average, implying that next time, there could be a heat exchanger that dissipates less heat to the surroundings.

1 Introduction

Plate heat exchangers have high efficiency, which is crucial since energy consumption and greenhouse gasses are increasing.¹ Heat exchangers help extract heat and place it someplace else, as heat always moves in the direction from a higher temperature to lower temperature. Plate heat exchangers can be made extremely small and have very broad applications since dissipating heat is nearly always a part of manufacturing. Some common industries are air conditioners, microstructure devices, and wastewater heat pumps.¹⁻³ Plate heat exchangers have a trade-off, being more cost heavy while having higher energy efficiency.¹ This expands to many other industries this product easily displacing heat allows chemical reactions to occur under optimal temperature conditions which can improve their reaction rates. This topic is relevant because it is very versatile and can be made large or compact for industry-level or microstructure devices. Heat exchangers started off in the food and air conditioning industry, but later moved on to chemical plants.² The heat exchanger can be used in sewage systems as it can recover lost thermal energy supplied from building, where the energy loss is reported at over 15%.² Heat exchangers have a heavy use in refrigeration and air conditioning systems, since it takes heat out of the system to cool, or into the system in a heater to increase thermal energy. Heat exchangers have classically used shell-and-tube methods which have a much simpler design. This can now be improved in recent years as designs are updated within heat exchanger networks to use plate heat exchangers. Plate heat exchangers are able to transfer heat more efficiently and able to transfer heat as low as 2°C.¹ Due to the more compact nature of the plate heat exchanger, they are also able to increase heat recovery or decrease energy consumption with smaller heat transfer area, which leads to the rise of heat exchangers working in microstructures.^{1,3} Micro-size heat exchangers have many benefits in electronics and biomedicine, but are limited by uncertainty with small scale systems in the thermal flow as there is no way to precisely measure the temperature distribution.³

2 Background

Since their invention in 1923, plate heat exchangers became popular in the dairy and paper/pulp industries due to hygienic reasons, since they are easily cleaned.⁴ Currently, plate heat exchangers are known for their wide application since they have a larger heat transfer area to volume ratio, lighter weight, design flexibility, and high thermal effectiveness, compared to other types of heat exchangers.⁵ The pattern of the plates is designed to maximize effective surface area of the plates, promote turbulence, and distribute the fluid flow. This is achieved due to the countercurrent flow and the design of the plates which allows the streams cross each other at an intersection causing turbulent flow and increasing the pressure drop.⁶ The geometry of the plates, especially the herringbone angle β influences the heat transfer properties. Experiments have been conducted for a range of herringbone angles varying from 20-65 degrees.⁵ One downside of a plate heat exchanger is fouling issues. This phenomenon happens when hard water is heated and calcium and bicarbonate ions precipitate onto the plates. This unwanted buildup “decreases the overall heat transfer coefficients and increases resistance to fluid flow”.⁶ It has been proven that a low herringbone angle, around 30, has a larger fouling resistance compared to plate with a higher angle of 63. The research on the fouling factor is limited and fouling factors correlations can not be widely applied. Empirical correlations of the Nusselt number and friction factor have been experimentally achieved, but lack applicability to all fluids and herringbone angles without specific experimental data. Research is also limited for different types of fluid and especially for 2 phase flow.⁴

In this study water is the working fluid on both sides of the plate. By using both hot and cold water reservoirs, and temperature measurements of the inlet and outlet of both streams the heat transfer is calculated. From the amount of heat transferred, correlations for the Nusselt number coefficients for a seven plate heat exchanger will be calculated.

3 Theory

From the known liquid flow rates, the heat transfer power for both sides can be calculated using the equation

$$Q = \frac{m_h * C_h(T_{H,i} - T_{H,0}) + m_c * C_c(T_{c,0} - T_{c,i})}{2} \quad (1)$$

where Q is the heat transfer power, m is the mass flow rate, c is the specific heat, and T is the temperature. The difference between the hot and side power is minimal since heat transferred out of the hot stream should in theory be transferred to the cold stream. With this assumption this equation gives an average for both the hot and cold sides. From this value of Q the heat transfer coefficient can be calculated using the equation

$$U = \frac{Q}{A_{proj} * \Delta T_{LMTD}} \quad (2)$$

where A_{proj} is the projected heat transfer area and ΔT_{LMTD} is the counter-flow log-mean temperature difference. The projected heat transfer area is defined as

$$A_{proj} = W_1 * L_2 \quad (3)$$

where W_1 is the outer plate width and L_2 is the inner plate length. The log mean temperature difference is defined as

$$\Delta T_{LMTD} = \frac{[(T_{h,in} - T_{c,out}) - (T_{h,out} - T_{c,in})]}{\ln[(T_{h,in} - T_{c,out}) - (T_{h,out} - T_{c,in})]} \quad (4)$$

For developing a heat transfer correlation for single-phase flow, the modified Wilson plot technique is adopted. The correlation can be written as follows

$$Nu = C_1 Re^{C_2} Pr^{1/3} \left(\frac{\mu}{\mu_w} \right)^{14} \quad (5)$$

The Reynolds number is defined as

$$Re = \frac{\rho V D_h}{\mu} \quad (6)$$

Where ρ is the density, D_h is the hydraulic diameter, which can be defined as

$$D_h = 2b \quad (7)$$

Where b is the corrugation depth, and V is the velocity inside the BPHE defined in the form:

$$V = \frac{m}{\rho n W_1 b} \quad (8)$$

Where n is the number of channels in parallel, and W_1 is the width of the plate.

The Nusselt number can be defined as

$$Nu = \frac{h D_h}{k} \quad (9)$$

where k is the thermal conductivity of the fluid.

4 Methods

This experiment used a hot and cold water reservoir, pumped into a counter-current heat exchanger with in and out flows for both hot and cold streams. 108 unique trials were run by varying combinations of hot and cold flow rates with increments by 2 Liters Per Minute, from 4LPM to 26LPM in random order to decrease error. This heat exchanger had 3 channels by utilizing 7 plates [Fig. 1](#). Each plate has a corrugation depth, shown in the figure as b , of 5.16 mm, plate thickness of 0.710 mm, outer plate width of 134 mm, inner plate length of 381 mm, and projected heat transfer area of 0.170 m² [Eq. \(3\)](#). The 4 steady state temperatures were measured from the inlet and outlet streams of each the hot and cold streams. Heat transfer coefficient and heat flow were determined from the temperatures and flow rates [Eqs. \(1\), \(2\) and \(4\)](#).

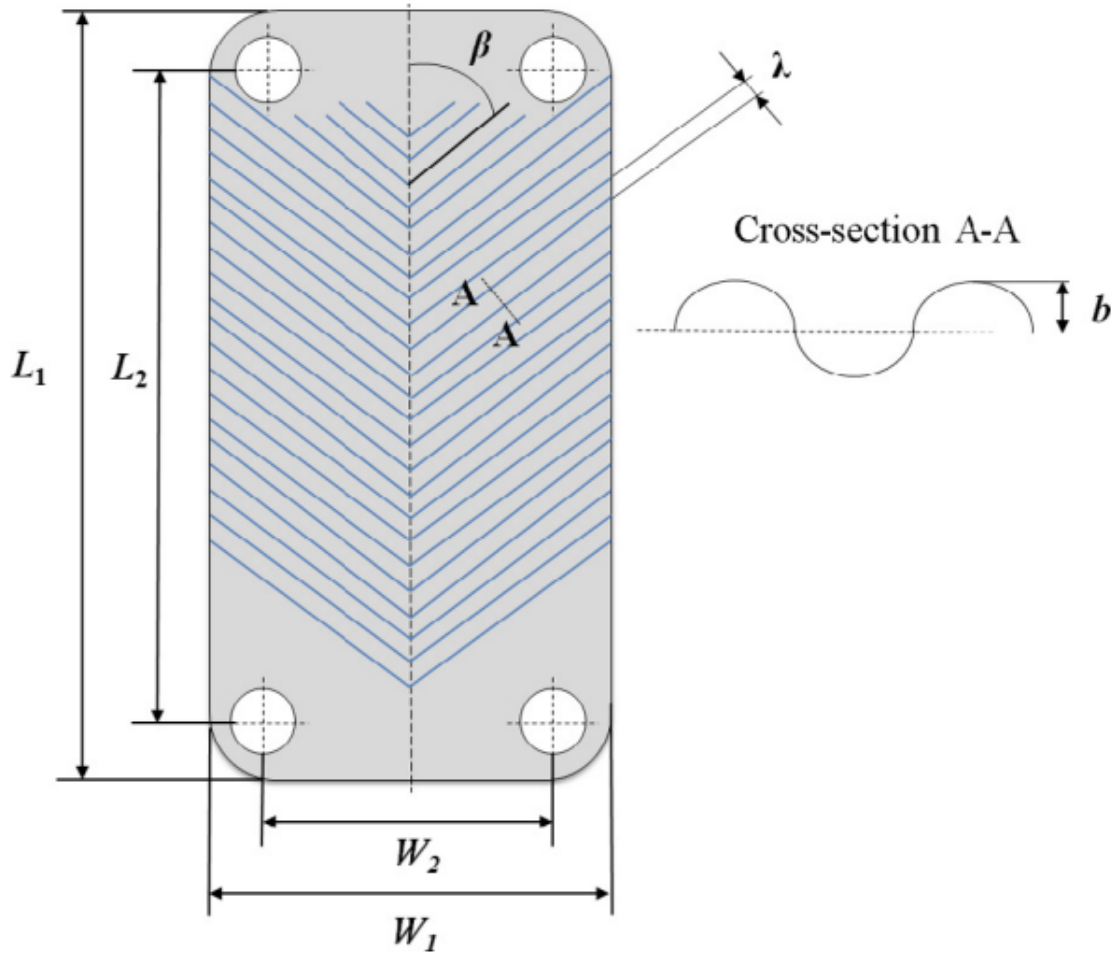


Figure 1: The schematic diagram of plate heat exchanger.

5 Results

5.1 Calculation of U from experiment

Initial and final temperatures of the hot and cold streams were tabulated from the experiment varying the flow rates of either stream, and from these values the measured values for the overall heat transfer coefficient U was derived. The heat overall transfer rate Q was determined as the average heat transfer rate between the two streams according to Equation 1. The temperatures at which the properties of the cold and hot streams were evaluated were calculated at the average of the inlet and outlet temperatures. The water specific heat, density, viscosity, and

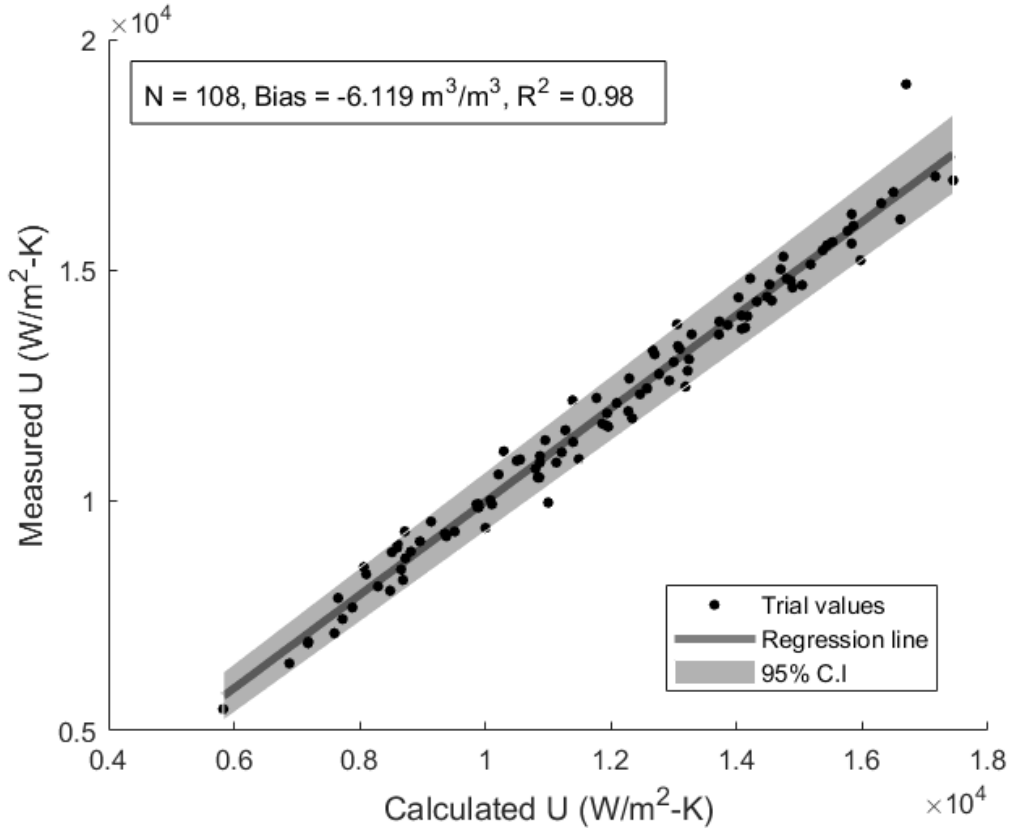


Figure 2: The calculated U was plotted against the measured U for 108 trials of random combinations of temperatures and flow rates of the hot and cold streams. Near identity was established by iteration of the coefficients of Equation 5.

thermal conductivity in the temperature range 20°C to 80°C accounted for using a second-order polynomial fit of tabulated data. The overall heat transfer coefficient U was determined using the log mean temperature difference according Equation 2.

5.2 Determination of the coefficients of the Nusselt number relation

The thermal conductivity of the plates was taken as that of carbon steel. The density, dynamic viscosity, heat capacity, and thermal conductivity of water were taken at the average temperature between the inlet and the outlet of each stream. The hydraulic diameter was used to define the Reynold and Nusselt numbers with accordance to Equations 5 and 6. The constants of Equation 5 were determined by iteration to agree with the experimental overall heat trans-

fer coefficient. The final values of the coefficients approaching the agreement between the theoretical and experimental values were determined to be $C_1 = 1.7868$ and $C_2 = 0.6853$. The highest Reynold number measured was 235, and the lowest was 34. This range of values is well within the application of Equation 6.

5.3 Calculating EBD

The energy balance deviation is defined as

$$EBD = \frac{Q_h - Q_c}{Q_{ave}}$$

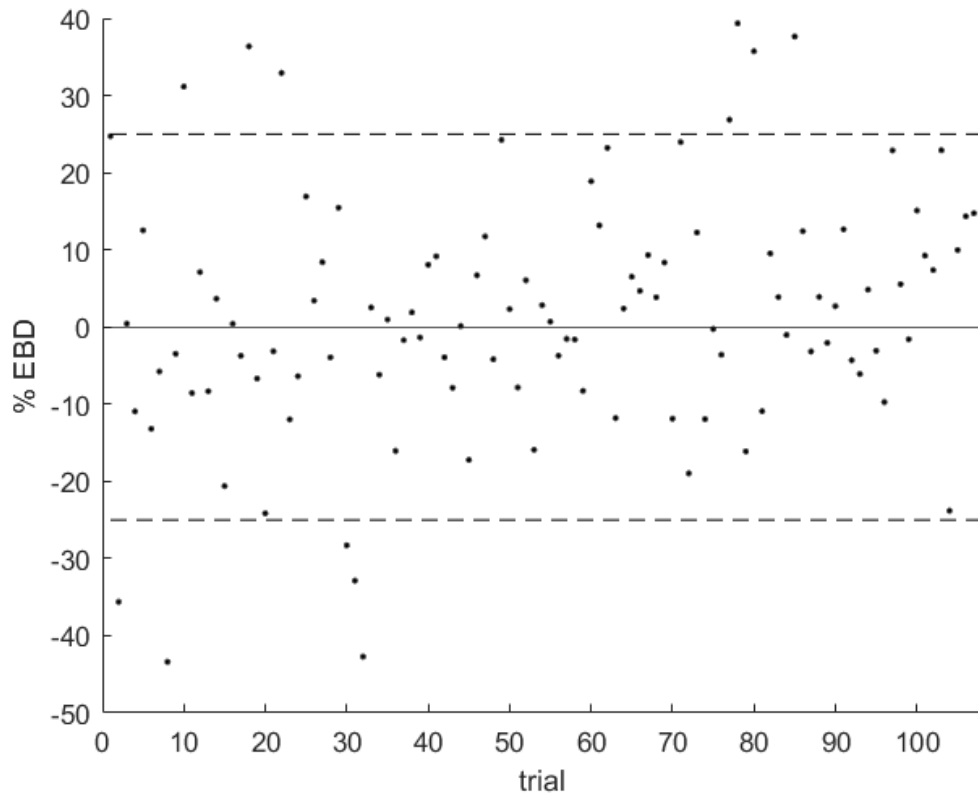


Figure 3: The EBD was plotted for each trial. 88.89% of the differences between the hot and cold flow rates fall between -25% and 25% of their average.

where Q_h and Q_c are the heat transfer rates of the hot stream and the cold stream respectively, and Q_{ave} is the average of Q_h and Q_c . The EBD shows the proximity of the experimental trial to ideal heat transfer between the two plates where no thermal energy is lost to the surroundings. 52.78% of the the EBD were found to be positive, and 47.22% of the EBD were found to be negative. The maximum EBD for a trial was found to be 39.34% while the minimum was found to be -43.39%. The average EBD was found to be 1.337%.

6 Discussion

Though we were able to find a satisfactory agreement between our fitted coefficients for the Nusselt number relation, the quality of the data is questionable due to magnitude of EBD values. The deviations from ideal heat transfer due to the loss and gain of heat to and from the surroundings is shown by the prevalence large values of the relative EBD. Heat loss and gain occurred at random as shown by the lack of trends in the data in Fig.3. This distribution of data shows the lack of control of the the direction of heat transfer as the plates were not insulated in the experiment. The fitted coefficients are applicable to only similar circumstances where the plates are not insulated and the likelihood of loss and gain of heat to and from the surroundings are equal as the average EBD was relatively low.

Compared to the experimental procedure from which ours is derivative, differences our lack of a water chiller and heater continuously running during the experiment which would have kept the streams at a constant temperature during the trials, our higher range of flow rates (0.066kg/s to 0.43kg/s) compared to 0.0378kg/s to 0.378kg/s, and our higher range of temperatures (66°C for the initial temperature of the hot stream, 21°C for the initial temperature of the cold stream) compared to 35°C and 5°C used in Yang et al.⁴ There is the possibility that the temperature of the plates equilibrated with the ambient temperature without hot and cold water at constant temperatures keeping the temperature of the plates near that of the respective streams. The temperature gradient, so to say the potential gradient of heat transfer

was smaller in the direction between the plates in our experiment. This difference may explain the greater loss of heat to the surroundings as the temperature gradient between the plates is smaller relative to that between the plates and the surroundings. The greater water flow rates and temperature differences indicated that greater heat transfer rates were achieved. The highest measured overall heat transfer coefficient in our experiment was $19000\text{Wm}^{-2}\text{K}^{-1}$ whereas in Yang et al. it was $3700\text{Wm}^{-2}\text{K}^{-1}$ for BPHE#6.⁴

7 Conclusions

In this experiment, the theoretical model matches the trend from the experimental values shown by Fig. 2, where a large majority the 108 trials fall under the 95% confidence interval. The theoretical values of overall heat transfer also match the experimental values shown by , where the combination of these two show these values are correlated with a model and correctly modeled. This proves the correlation of determining Nusselt's number from Prandtl's and Reynold's numbers was accurate, with the theoretical model, Eq. (5) with $C_1 = 1.7868$ and $C_2 = 0.6853$. The EBD has large outliers and has overall wide distribution due to the thermal energy being lost to the surroundings of the plate heat exchanger. In the future, we could potentially create an insulated jacket environment around the heat exchanger to prevent thermal energy from exiting the system.

Bibliography

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- (2) Culha, O. Heat exchanger applications in wastewater source heat pumps for buildings: A key review. *Energ. Buildings*. **2015**, 104, 215–232.

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- (4) J. Yang, e. a. Heat transfer correlations for single-phase flow in plate heat exchangers based on experimental data. *App. Th. Eng.* **2017**, 113, 1547–1557.
- (5) R.L. Pradhan, e. a. Review of Nusselt Number Correlation for Single Phase Fluid Flow though a Plate Heat Exchanger to Develop C Code Application Software. *IOSR-JMCE* **2013**, 4, 1–8.
- (6) Cremaschi, L. Effect of Condensation Temperature and Water Quality on Fouling of Brazed-Plate Heat Exchangers. *ASHRAE Transactions* **2012**, 12, 1086–1100.

Appendix

1. First main reference

- Author(s): Xu, Kevin, et al.
- Year published: 2019
- Journal name: Computer Aided Process Engineering
- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: Replacing shell-and-tube heat exchangers with plate heat exchangers, allows higher efficiency with smaller heat transfer area.

2. Second main reference

- Author(s): Culha, O., et al.
- Year published: 2015
- Journal name: Energy and Buildings
- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: Heat exchangers in waste water systems can retrieve 90% of disappeared energy from heat along the sewage channel.

3. Third main reference

- Author(s): Brandner J. J., et al.
- Year published: 2006
- Journal name: Proceedings of the 4th International Conference on Nanochannels, Microchannels, and Minichannels
- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: Heat exchangers can be used at micro reaction engineering level with integrated channels.

4. Fourth main reference

- Author(s): R.L. Pradhan, et al.
- Year published: 2017
- Journal name: IOSR-JMCE
- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: Developed a code application software that can give the heat transfer coefficients for a plate heat exchanger given geometry patterns and flow conditions. Nusselt number correlations can be found for beta angles ranging from 20-65 and water as the fluid.

5. Fifth main reference

- Author(s): J. Yang, et al.
- Year published: 2017
- Journal name: App. Th. Eng.
- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: For plate heat exchangers the geometric dimensions, especially the herringbone angle affect the heat transfer the most.
 - (b) Second accomplishment: The correlations found for single-phase heat transfer can calculate the heat transfer performance for a large range of plate heat exchangers with an accuracy of 50

6. Sixth main reference

- Author(s): L. Cremaschi, et al.
- Year published: 2019
- Journal name: ASHRAE Transactions

- 1-3 major accomplishments of this paper:
 - (a) First accomplishment: Found that heat flux and wall plate surface temperature were responsible for fouling of brazed plate heat exchangers and that plates with soft corrugation angles were more sensitive to these factors.
 - (b) Second accomplishment: Found that the measured pressure drop in already fouled plates were 10-11 percent higher than in plates that were clean.
 - (c) Third accomplishment: The water quality had a large impact on fouling, high-fouling potential water increased the fouling resistance by as much as 69 percent.